Alexandre Mosquetti Ramalho de Castro

Solar PV & Electrical Design Engineer • Utility-Scale • LV/MV • EV Charging

Location: Fortaleza, Brazil • Open to relocation • Eligible: EU Blue Card · CSEP (IE) · PAC (ES)

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PROFESSIONAL SUMMARY

Electrical Engineer | +7 years' experience

Electrical Engineer with over 7 years of experience delivering utility-scale solar PV, LV/MV electrical systems, building-services engineering and EV charging infrastructure. I work across the full project lifecycle, from PVsyst simulations and layout optimization to complete IFC documentation and drawing packages: single-line and three-line diagrams, protection schemes, AC/DC routing & diagrams, cable sizing, BoMs and utility connection packages, across residential, commercial, and industrial projects. My work also includes technical coordination and integration of auxiliary low-current systems, such as CCTV, monitoring, and communication infrastructure, as part of complete plant designs. Fluent in English (written and spoken).

To date, I have designed and delivered 149 electrical projects, 99 of them solar PV. 67 are full IFC packages totaling 53+ MWp DC across 12 Brazilian states, and 41 obtained utility grid-connection approval (36.9 MWp, 11 states, 10 distribution utilities). I have also designed 50 EV charging systems (1.69 MW AC, 182 charging points) and simulated 140+ MWp in PVsyst. More recently, I have applied the same engineering discipline to digitalize and automate utility operational workflows, building Python/SQL/Playwright-based tools now used daily by a 7-analyst team.

Proficient in AutoCAD (10+ years), Revit (intermediate), PVsyst, PVCase (evaluation/testing), SketchUp, MS Office and Excel automation. Also proficient in Python, SQL, Power BI, Salesforce, Power Automate and Playwright for data analysis, reporting and process automation.

SKILLS

Engineering & Design: Utility-scale Solar PV Design • IFC documentation & Technical Reports • BESS Design • LV/MV Power Distribution (load analysis, cable sizing, grounding, protective device selection) • PV System Performance Analysis (simulations, shading, yield) • Layout Optimization & Terrain-Based Design • EV Charging Systems (EU/IEC-compliant) • Auxiliary Systems Design (CCTV, monitoring and communication infrastructure) • Bills of Materials (BoMs) & Construction Packages • QA/QC & Compliance Verification • 2D/3D CAD Drafting • Engineering Process Standardization & Automated Spreadsheets • Process Automation & Data Analytics

Software & Tools: AutoCAD • Revit (intermediate – project ready) • PVsyst • PVCase (evaluation/testing – terrain & layout workflows) • SketchUp • Excel (automation, dashboards, engineering calculators) • Microsoft Office • AI-Assisted Tools (documentation generation, QA/QC, template automation, technical writing optimization) • Python (automation & data extraction) • SQL • Power BI • Salesforce • Power Automate • Playwright

Standards: IEC 60364 (NBR 5410) • IEC 61936-1 (NBR 14039) • IEC 62305 (NBR 5419) • IEC 62548 (NBR 16690) • IEC 60364-7-722 / IEC 61851 (NBR 17019) • IEC 61727 (NBR 16149) • IEC 62116 • IEC 62446-1 (NBR 16274) • IEC 60287 (NBR 11301) • IEC TR 63227 • ANEEL REN 1.000/2021 • PRODIST Module 3 • Law 14.300/2022

Languages: Portuguese — Native | English — Fluent | Spanish — Pre-Intermediate (A2) | Italian — Basic

EDUCATION

Bachelor's Degree in Electrical Engineering

Federal University of Ceará (UFC) — Fortaleza, Brazil

Undergraduate thesis (2024): Residential BESS design — grid-tied PV with lithium-ion storage, PVsyst with 3D SketchUp shading, three dispatch strategies compared under Brazil's Law 14.300 (Fio B).

PROFESSIONAL EXPERIENCE

ENEL DISTRIBUIÇÃO CEARÁ (via NTT Data)

Senior BPO Consultant (Process & Data Automation) — Grid Connections **Apr 2026 – Present**

Fortaleza, Brazil

- Apply an engineering, systems-thinking approach to digitize and automate the grid-connection request workflow for large commercial, industrial, and distributed-generation customers at Enel Distribuição Ceará, one of Brazil's largest electric utilities. Manage the connection process end-to-end, from request intake through energization of the customer's connection point, ensuring compliance with ANEEL's national regulatory connection-time standards (REN 1.000). Built Python, SQL, and Playwright-based automation pipelines, replacing scattered manual workflows across the team.

Key Achievements

- Engineered a Python + Playwright automation that processes utility connection orders in Salesforce at a rate of 1 order every 2 min 42 sec, running unattended around the clock, versus a manual capacity of ~100 orders/day per full-time analyst, freeing the team to focus on higher-value work.
- Designed and maintain an operations dashboard that consolidates 12 disparate data sources (Synergia, SharePoint, the utility's GD connection portal, corporate BI tools, and automated email ingestion via Power Automate) into a single system of record, cutting data-retrieval and analysis time by ~90% for a team of 7 analysts, while consolidating the team's day-to-day tools into one place.
- Built an AI-assisted data pipeline that extracted and structured the full communication history of 451 connection-request records in 1.5 hours, then ran automated qualitative and quantitative analysis to identify the root causes behind request rejections, as well as customer behavior following a rejection, insights that directly informed process-improvement proposals to leadership.

- Developed 25+ automation scripts and tools (Python, VBA/macros, Power Automate flows) for data extraction, transformation, and cross-referencing across the team's systems, eliminating repetitive manual work and supporting leadership-level diagnostics and reporting.

GESIN SOLAR

Solar PV Electrical Project Engineer **Apr 2024 – Mar 2026**

Fortaleza, Brazil

- Delivered complete IFC and utility grid-connection projects for PV plants ranging from 100 kW to 3.3 MWp, covering optimized layout design, full electrical dimensioning (DC/AC cable sizing, voltage-drop analysis, protection schemes, grounding/earthing), auxiliary systems integration (CCTV, monitoring and communication infrastructure), and construction-ready documentation. Managed utility resubmission workflows and PVsyst performance simulations across the full project portfolio.

Key Achievements

- Designed and delivered 10 PV projects totaling 7.23 MWp DC, from commercial rooftops to a 3.3 MWp ground-mount plant; 6.74 MWp approved by the utility, including plants of 3.3 MWp and 1.99 MWp and four co-located sites totaling 1.45 MWp.
- Optimized plant layout inherited from a previous engineer, reducing cable requirements and material costs affected by the layout change by roughly 30%, directly improving project BoM and construction budget.
- Resubmitted and secured utility approval for all 6 plants following design revisions, achieving zero rejection responses from the local grid operator across all reapproval submissions.
- Delivered two zero-export PV systems (100 kW and 240 kW · 0.40 MWp), cutting the client's energy bill by an estimated BRL 297,000 a year.

ION MOBILIDADE

EV Charging Systems Designer (concurrent role) **Apr 2024 – Dec 2025**

Fortaleza, Brazil

- Designed and delivered EV charging infrastructure projects for residential, commercial, and condominium buildings, managing full project scope from site survey and load analysis to LV distribution design, protective device coordination, and compliance documentation. Applied EU/IEC standards throughout and developed company-wide documentation frameworks adopted across all project types.

Key Achievements

- Designed 50 EV charging systems totaling 1.69 MW AC and 182 charging points, across 12 commercial sites and 20 condominiums; the largest single project is a 103-unit condominium totaling 762 kW.
- Achieved 90% first-submission approval rate across all projects.
- Accelerated technical documentation production by 40–60% through AI-enhanced drafting and QA/QC workflows, reducing revision cycles by over 50%.
- Standardized 6 core technical documents from scratch (site survey checklist, installation manual, user manual, compliance verification report, technical memorandum, and liability declaration), adopted company-wide across all project types.

ENAUT Soluções em Instalações

Electrical Projects Consultant (concurrent role) **Dec 2024 – Dec 2025**

Fortaleza, Brazil

- Reviewed third-party electrical designs and developed accurate Bills of Materials for tendering submissions, supporting the company's bid process for construction contracts across industrial, healthcare, and commercial sectors. Ensured multidisciplinary project compatibility (civil, mechanical, hydraulic, and electrical) prior to execution.

Key Achievements

- Reviewed and produced BoMs for electrical projects across industrial facilities, hospitals, and commercial buildings, directly supporting competitive tender submissions.
- Led electrical design review and full update of a hospital expansion project incorporating an MRI suite, ensuring compliance, spatial compatibility, and safe integration of high-interference medical equipment.
- Prevented execution-stage rework across all reviewed projects through rigorous multidisciplinary clash detection and compliance verification.

Independent Solar Design Engineer

Photovoltaic Project Engineer **May 2024 – Aug 2024**

Fortaleza, Brazil

- Delivered PV design and energy simulation services as an independent consultant, supporting utility-scale project development for a solar engineering firm.

Key Achievements

- Designed a 1.25 MW (AC) solar plant in Rio Grande do Sul, delivering full IFC package: drawings, electrical documentation, layout, and PVsyst performance studies.
- Performed 21 PVsyst simulations for 9 utility-scale plants totaling 22.6 MW (AC) within a one-week deadline, covering configuration optimization, shading assessment, and yield calculations.
- Conducted advanced energy simulation for a 60+ MWp utility-scale plant co-located with a wind farm, using full 3D terrain modelling in PVsyst to quantify shading losses from wind turbines across multiple string configurations and production horizons (year 1, 10, and 35); outputs were decisive in the client's final layout selection.

ECO Soluções em Energia

Engineering Supervisor **Jun 2021 – Dec 2023**

Fortaleza, Brazil

- Led the electrical engineering team at ECO Soluções, overseeing a multidisciplinary group of 2 civil engineers and 4 electrical designer engineers across the full delivery of utility-scale solar PV projects up to 5 MW. Responsible for third-party project review, cross-disciplinary workflow coordination, technical quality assurance, and strategic input on large-scale project budgeting. Served as Key Electrical User during the company's formal PVCase evaluation phase.

Key Achievements

- Directly delivered 32 solar PV projects (IFC and/or grid-connection packages) during tenure. Across my time at ECO: 78 PV projects, 55 IFC packages (45.5 MWp DC) and 33 grid-connection approvals (29.5 MWp) across 11 states and 10 distribution utilities.
- Contributed to the electrical sizing (LV/MV AC and DC cabling, conduit sizing) of three utility-scale centralized-generation budgets totaling 274 MWp.
- Reviewed 10+ third-party electromechanical projects for technical compliance and constructability, directly taking over and completing final revisions on multiple projects where third-party delays were jeopardizing delivery schedules.
- Acted as Key Electrical User for PVCase evaluation, leading technical assessment in English with the software's support and commercial teams, applying the tool across 3 utility-scale projects with challenging terrain conditions, and provided the internal technical recommendation on adopting the tool.
- Evolved and formalized the multidisciplinary drawing workflow established during the EPC coordination phase, improving coordination between civil and electromechanical teams as the department scaled.

Previous Roles at ECO Soluções em Energia:

Coordinator – EPC Department **Feb 2020 – May 2021**

- Joined as a founding member of the newly established EPC department, helping build the sector from the ground up alongside the Engineering Manager. Coordinated a team of 2 designers across full IFC and grid-connection project development for solar PV plants up to 2.5 MW, overseeing documentation quality, workflow standardization, and budget estimation for projects up to 40 MW.

Key Achievements

- Co-founded and structured the EPC department from inception, defining document standards, drawing conventions, and project delivery workflows in direct collaboration with the Chief Electrical Engineer, establishing the operational framework adopted by the growing team.
- Delivered 16 solar PV projects (IFC and/or grid-connection packages) during the period.
- Supervised budget estimations for utility-scale plants up to 40 MW, providing technical cost input to support competitive project development.
- Led a 6-week cross-functional task force offsite to fast-track submissions for 14 utility-scale PV plants simultaneously, coordinating electrical, civil, and environmental teams to manage both grid-connection submissions and environmental licensing support packages.
- Contributed to significant department growth from kickoff to a fully operational multi-disciplinary team within 15 months, leading to promotion to Engineering Supervisor.

Electrical Designer II **Sep 2019 – Feb 2020**

- Produced electrical and electromechanical IFC drawings for solar PV projects up to 300 kW, including routing layouts for power, grounding, and auxiliary low-current systems. Mostly Residential and Commercial projects. Prepared grid-connection documentation and technical packages for utility submission and supported engineering coordination meetings.

Key Achievements

- Accountable for the technical development of 21 PV projects (accumulated since the internship period), with direct involvement in electrical design and utility grid-connection approval workflows.
- Led electrical design of large-scale PV projects for major regional dealerships and distributors, marking the transition to higher-complexity utility-scale work.
- Introduced SketchUp 3D modelling into the design workflow: building detailed site models including surrounding shading elements for export to PVsyst, improving simulation accuracy and deliverable quality.

Engineering Intern **Oct 2018 – Aug 2019**

- Supported senior engineers in preparing electrical and electromechanical drawings for solar PV projects up to 75 kW, contributing to IFC and utility grid-connection packages, BoM generation, and project documentation, mostly Residential and Commercial. Rapidly developed proficiency in AutoCAD and Excel, building a strong technical foundation in PV design workflows.

Key Achievements

- Recognized for fast-track technical development. Promoted directly to Designer II (skipping Designer I) based on design quality and speed of learning demonstrated during the internship period.

OTHER COURSES

2026 | Strategic Training in BESS (10 h) — Silvia Resende
2020 | Commissioning of grid-connected photovoltaic systems (16 hrs) – Elekt Solar
2016 – 2018 | MATLAB, Smart Lighting, MS Project

ADDITIONAL INFORMATION

Eligible to apply for the EU Blue Card, the Critical Skills Employment Permit (CSEP · Ireland) or the PAC highly qualified professional permit (Spain). Open to relocation and remote positions.